CS2600

THE UNIVERSITY OF WARWICK
Level 5 Open Book Assessment — 2 hours
Department of Computer Science, April 2019/20
CS260 Algorithms

Instructions

1. Read all instructions carefully — and read through the entire paper at least once before you start writing.

2. There are 2 questions in Section A and 3 questions in Section B.
   Answer **THREE** questions: **ONE** from Section A and **TWO** from Section B.

3. All questions will carry the same number of marks unless otherwise stated.

4. You should handwrite your answers either with paper and pen or using an electronic device with a stylus (unless you have special arrangements for exams which allow the use of a computer). Start each question on a new page and clearly mark each page with the page number, your student id and the question number.
   Handwritten notes must be scanned or photographed and all individual solutions should (if you possibly can) be collated into a single PDF with pages in the correct order.
   You must upload two files to the AEP: your PDF of solutions and a completed cover sheet.
   You must click **FINALISE** to complete the submission process. After you have done so you will not be able to upload anything further.

5. Please ensure that all your handwritten answers are written legibly, preferably in dark blue or black ink. If you use a pencil ensure that it is not too faint to be captured by a scan or photograph.

6. Please check the legibility of your final submission before uploading. It is your responsibility to ensure that your work can be read.

7. You are allowed to access module materials, notes, resources, references and the internet during the assessment.

8. You should not try to communicate with any other candidate during the assessment period or seek assistance from anyone else in completing your answers. The Computer

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Science Department expects the conduct of all students taking this assessment to conform to the stated requirements. Measures will be in operation to check for possible misconduct. These will include the use of similarity detection tools and the right to require live interviews with selected students following the assessment.

9. By starting this assessment you are declaring yourself fit to undertake it. You are expected to make a reasonable attempt at the assessment by answering the questions in the paper.

Please note that:

— You must have completed and uploaded your assessment before the 24 hour assessment window closes.

— You have an additional 45 minutes beyond the stated length of the paper to allow for downloading and uploading the assessment, your files and technical delays.

— For further details you should refer to the AEP documentation.

Use the AEP to seek advice immediately if during the assessment period:

* you cannot access the online assessment;

* you believe you have been given access to the wrong online assessment.

Please note that technical support is only available between 9AM and 5PM (BST). Invigilator support will be also be available (via the AEP) between 9AM and 5PM (BST).

Notify `comp-sci@dcs.warwick.ac.uk` as soon as possible if you cannot complete your assessment because:

- you lose your internet connection;
- your device fails;
- you become unwell and are unable to continue;
- you are affected by circumstances beyond your control (e.g. fire alarm).

Please note that this is for notification purposes, it is not a help line.

Your assessment starts on the next page.

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Section A

Answer **ONE** question from this section.

1. (a) **[6 points]** Suppose the radix sort algorithm is run on the following array of 3-digit positive decimal integers:

   137 991 134 829 235 989 250 15 223

   Draw a table showing the contents of the array after each iteration of the main loop of the radix sort.

   (b) **[8 points]** Does there exist a decision tree for sorting 3-element arrays that has fewer than 6 leaves? Justify your answer.

   (c) **[6 points]** Suppose that the Quicksort algorithm uses a pivot selection procedure which takes time $O(n)$, where $n = r - \ell + 1$ is the number of elements in the array. Is it true that every such version of Quicksort runs in time $O(n^2)$? Give a proof or a counterexample.

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2. (a) **[4 points]** For both of the following algorithms, specify what is given as input, what is the required output, and what is the worst-case running time: (i) Dijkstra's algorithm, (ii) Bellman–Ford's algorithm.

   (b) **[8 points]** Suppose a connected undirected weighted graph $G = (V, E)$ with positive weights has a *unique* minimum spanning tree T. Is it true that the path between every two distinct vertices in T is always a shortest path between these vertices in G? Give a proof or a counterexample.

   (c) **[8 points]** Consider the task of searching a sorted array $A[1..n]$ for a given element x, a task usually performed by binary search in time $O(\log n)$. Show that any algorithm that accesses the array only via comparisons (that is, by asking questions of the form "is $A[i] \leq z$?" and "is $A[i] \geq z$?") must take $\Omega(\log n)$ steps.

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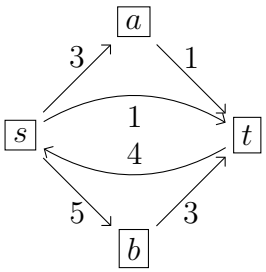
Section B

Answer **TWO** questions from this section.

3. (a) **[4 points]** Give the definition of *linear extension* of a directed graph.

   (b) **[7 points]** Give an instance of the set cover problem with universe of size ≤ 30 and 6 sets such that the greedy algorithm produces a cover twice the size of an optimal cover.

Consider the flow network $\mathcal{N}$ shown below.



(c) **[6 points]** Find a maximum-value (s,t)-flow in $\mathcal{N}$. What is the value of this flow?

(d) **[5 points]** Prove that for every (s,t)-flow f and every (s,t)-cut (L,R) in $\mathcal{N}$ the following equality holds:

$$\mathrm{val}(f) = \sum_{uv\colon\, u\in L, v\in R} f(uv) - \sum_{uv\colon\, u\in R, v\in L} f(uv)$$

(e) **[4 points]** Consider four sequences of integers:

$$\mathcal{A}: 0,0,1,1; \qquad \mathcal{B}: 0,0,0,1; \qquad \mathcal{C}: 0,1,1,1; \qquad \mathcal{D}: 2,0,1,0.$$

What is the length of the longest common subsequence of:
(i) $\mathcal{A}$ and $\mathcal{B}$? (ii) $\mathcal{A}$ and $\mathcal{C}$? (iii) $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$? (iv) $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, $\mathcal{D}$?

(f) **[2 points]** An $n \times n$ table of integers F is called *flat* if, for some k, its first row contains k zeroes and $n-k$ ones, in this order; in other words if $F[1][j]$ is equal to 0 when $1 \leq j \leq k$ and equal to 1 when $k+1 \leq j \leq n$.

Draw a 4×4 table of integers in which the contents of the first, second, third, and fourth rows are the sequences $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$ from part (e). Is this table flat?

(g) **[12 points]** Give a polynomial-time algorithm that, given an $n \times n$ flat table of integers, finds the length of the longest common subsequence of its n rows. Justify the correctness of your algorithm and prove that it runs in polynomial time.

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4. (a) **[5 points]** Suppose the Bellman–Ford algorithm is run on the directed graph with vertices $\{a, b, c, d, e\}$ and the following six edges, which the algorithm orders as $ad < ba < bc < be < dc < ed$. The weights of the edges are $w(ad) = 1$, $w(ba) = -2$, $w(bc) = 4$, $w(be) = 1$, $w(dc) = 1$, $w(ed) = 2$.

   Let the initial vertex be b. What is the sequence of edge relaxations that the algorithm performs?

   Draw a table showing the intermediate distance values of all the nodes *after each relaxation of the edge ed*.

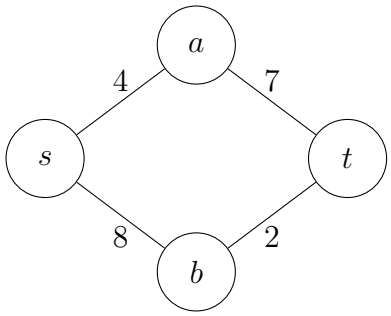
   (b) **[5 points]** Let $G = (V, E)$ be an undirected graph with weights $w\colon E \to \mathbb{R}$. Let $e \in E$ be an edge of maximum weight. Suppose T is a spanning tree of G that has the maximum weight among all spanning trees of G. Prove that, if e is the only edge of G with weight $w(e)$, then T must contain e.

   (c) **[6 points]** Suppose a nondecreasing function $f\colon \mathbb{Z}_{\geq 0} \to \mathbb{Z}_{\geq 0}$ satisfies the inequality $f(n) \leq \frac{2}{n-1} \cdot (n^2 + \sum_{i=1}^{n-1} f(i))$ for all $n \geq 2$. Prove that $f(n) = O(n \log n)$.

Let $G = (V, E)$ be an undirected graph with edge weights $w\colon E \to \mathbb{R}_{>0}$, and let s and t be two distinct vertices in V. Suppose that uv is not an edge of G, i.e., $uv \notin E$, but $u, v \in V$ and $u \neq v$.

For an arbitrary real number $q > 0$, denote by $G(q)$ the graph G modified as follows: the edge uv is added with weight q, and all other edges and vertices are as in G. Denote by $D(q)$ the length of the shortest path from s to t in $G(q)$.

(d) **[6 points]** For the weighted graph G depicted below, with w, s, and t as shown and with $uv = ab$, draw the graphs $G(2)$, $G(4)$, and $G(6)$.



(e) **[6 points]** For the example from part (d), compute the values $D(2)$, $D(4)$, and $D(6)$.

(f) **[12 points]** Give a polynomial-time algorithm that, given G, w, s, t, u, v, and a sequence of positive real numbers $q_1, \ldots, q_k$, computes the values $D(q_1), \ldots, D(q_k)$.

Justify the correctness of your algorithm and analyse its worst-case running time. For full credit, your algorithm should run in time $O((|V| + |E|) \log |V| + k)$.

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5. (a) **[3 points]** Give the definition of *NP-hardness*.

   (b) **[15 points]** For each of the following statements, indicate whether it is true or false. (Justification is not required. You will be awarded 3 points for each correct answer and -3 points for each incorrect or missing answer. Should the sum of the awarded points be negative, you will receive 0 points for this part of the question.)

   i. A decision problem is NP-complete if and only if there is a polynomial-time reduction from SAT to it.

   ii. A decision problem is NP-hard if and only if there is a polynomial-time reduction from it to SAT.

   iii. As far as we know, it is possible that the minimum spanning tree problem has no polynomial-time algorithm.

   iv. The Ford–Fulkerson algorithm for the maximum flow problem is not a polynomial-time algorithm.

   v. For every two problems $X, Y \in \mathrm{P}$, we have $X \leq Y$ and $Y \leq X$.

   (c) **[8 points]** Prove that the following *Two Disjoint Paths* problem belongs to the complexity class NP:
   Input: directed graph $G = (V, E)$, vertices $s_1, t_1, s_2, t_2 \in V$.
   Output: does G contain two paths $s_1 \to^* t_1$ and $s_2 \to^* t_2$ that do not share any vertex with each other?

   (d) **[8 points]** Prove that the following *Hitting Rows* problem is NP-hard:
   Input: matrix $A \in \{0,1\}^{m \times n}$ and integer $k \leq m$.
   Output: does there exist a collection of k rows of A such that, for each column of A, at least one of these k rows has a 1 in this column?

   (e) **[6 points]** Give a positive instance of the *Hitting Rows* problem with $m = n = 3$ and $k = 2$. Does there exist a negative instance of this problem with $m = n = k = 3$? Justify your answer.

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